THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Yes	Two sides together must be longer than the third.
2	**(b)** 70°	The three angles add up to 180°.
3	**(c)** equilateral	Recall the definition.
4	**(a)** 95°	Exterior angle = sum of the two interior opposite angles.
5	**(d)** 70°	Base angles are equal and the three angles add up to 180°.
6	**(b)** 21	Add the three sides.
7	**(c)** SSS	Compare the matching parts.
8	**(b)** 65°	The two acute angles add up to 90°.
9	**(b)** No	Two sides together must be longer than the third.
10	**(c)** 70°	The three angles add up to 180°.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Two shorter sides: $6 + 8 = 14$; Compare with the longest side 10 (1); Yes, 14 > 10 (1).	1+1
12	Third $= 180^\circ - (48^\circ + 72^\circ)$; $= 60^\circ$ (1); The triangle is acute-angled (1).	1+1
13	$\angle C = 180^\circ - 105^\circ = 75^\circ$; $\angle ACD = 180^\circ - 75^\circ$ (1); $= 105^\circ$ (equal to $\angle A + \angle B$) (1).	1+1
14	Two shorter sides: $3 + 5 = 8$; Compare with the longest side 9 (1); No, 8 is not greater than 9 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total parts $= 6$; one part $= 180^\circ \div 6 = 30^\circ$ (1); Angles 30°, 60°, 90° (1); Check: 180° (1).	1+1+1
16	Base angles are equal; their sum $= 180^\circ - 30^\circ = 150^\circ$ (1); Each base angle $= 75^\circ$ (1); Check: the three angles add up to 180° (1).	1+1+1
17	Exterior = sum of interior opposite angles (1); Other $= 120^\circ - 45^\circ = 75^\circ$ (1); Adjacent interior angle $= 180^\circ - 120^\circ = 60^\circ$ (1).	1+1+1
	OR: Total parts $= 9$; one part $= 180^\circ \div 9 = 20^\circ$ (1); Angles 40°, 60°, 80° (1); Check: 180° (1).	1+1+1
18	Both base angles are 70°, sum $= 140^\circ$ (1); Apex angle $= 180^\circ - 140^\circ$ (1); $= 40^\circ$ (1).	1+1+1
	OR: Exterior = sum of interior opposite angles (1); Other $= 110^\circ - 60^\circ = 50^\circ$ (1); Adjacent interior angle $= 180^\circ - 110^\circ = 70^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$5 + 5 = 10 > 8$, so the triangle exists (1); It is isosceles (1); Perimeter $= 5 + 5 + 8$ (1); $= 18$ cm (1); Final statement (1).	1+1+1+1+1
	OR: $\angle C = 180^\circ - 40^\circ - 65^\circ = 75^\circ$ (1); Classification: acute-angled (1); Exterior angle at C $= 180^\circ - 75^\circ$ (1); $= 105^\circ$ (1); It equals $\angle A + \angle B = 40^\circ + 65^\circ$ (1).	1+1+1+1+1
20	$6 + 8 = 14 > 10$, so the triangle exists (1); It is scalene (1); Perimeter $= 6 + 8 + 10$ (1); $= 24$ cm (1); Final statement (1).	1+1+1+1+1
	OR: $\angle C = 180^\circ - 50^\circ - 40^\circ = 90^\circ$ (1); Classification: right-angled (1); Exterior angle at C $= 180^\circ - 90^\circ$ (1); $= 90^\circ$ (1); It equals $\angle A + \angle B = 50^\circ + 40^\circ$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** isosceles	Recall the definition.
2	**(b)** 100°	Exterior angle = sum of the two interior opposite angles.
3	**(b)** 50°	Base angles are equal and the three angles add up to 180°.
4	**(c)** 23	Add the three sides.
5	**(c)** SAS	Compare the matching parts.
6	**(c)** 52°	The two acute angles add up to 90°.
7	**(d)** Yes	Two sides together must be longer than the third.
8	**(a)** 62°	The three angles add up to 180°.
9	**(d)** scalene	Recall the definition.
10	**(b)** 115°	Exterior angle = sum of the two interior opposite angles.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Third $= 180^\circ - (90^\circ + 35^\circ); = 55^\circ$ (1); The triangle is right-angled (1).	1+1
12	$\angle C = 180^\circ - 115^\circ = 65^\circ$; $\angle ACD = 180^\circ - 65^\circ$ (1); $= 115^\circ$ (equal to $\angle A + \angle B$) (1).	1+1
13	Two shorter sides: $7 + 7 = 14$; Compare with the longest side 12 (1); Yes, 14 > 12 (1).	1+1
14	Third $= 180^\circ - (110^\circ + 25^\circ); = 45^\circ$ (1); The triangle is obtuse-angled (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total parts $= 4$; one part $= 180^\circ \div 4 = 45^\circ$ (1); Angles 45°, 45°, 90° (1); Check: 180° (1).	1+1+1
16	Base angles are equal; their sum $= 180^\circ - 110^\circ = 70^\circ$ (1); Each base angle $= 35^\circ$ (1); Check: the three angles add up to 180° (1).	1+1+1
17	Exterior = sum of interior opposite angles (1); Other $= 135^\circ - 80^\circ = 55^\circ$ (1); Adjacent interior angle $= 180^\circ - 135^\circ = 45^\circ$ (1).	1+1+1
	OR: Total parts $= 12$; one part $= 180^\circ \div 12 = 15^\circ$ (1); Angles 45°, 60°, 75° (1); Check: 180° (1).	1+1+1
18	Both base angles are 50°, sum $= 100^\circ$ (1); Apex angle $= 180^\circ - 100^\circ$ (1); $= 80^\circ$ (1).	1+1+1
	OR: Exterior = sum of interior opposite angles (1); Other $= 100^\circ - 35^\circ = 65^\circ$ (1); Adjacent interior angle $= 180^\circ - 100^\circ = 80^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$9 + 9 = 18 > 9$, so the triangle exists (1); It is equilateral (1); Perimeter $= 9 + 9 + 9$ (1); $= 27$ cm (1); Final statement (1).	1+1+1+1+1
	OR: $\angle C = 180^\circ - 30^\circ - 50^\circ = 100^\circ$ (1); Classification: obtuse-angled (1); Exterior angle at C $= 180^\circ - 100^\circ$ (1); $= 80^\circ$ (1); It equals $\angle A + \angle B = 30^\circ + 50^\circ$ (1).	1+1+1+1+1
20	$7 + 9 = 16 > 12$, so the triangle exists (1); It is scalene (1); Perimeter $= 7 + 9 + 12$ (1); $= 28$ cm (1); Final statement (1).	1+1+1+1+1
	OR: $\angle C = 180^\circ - 35^\circ - 105^\circ = 40^\circ$ (1); Classification: obtuse-angled (1); Exterior angle at C $= 180^\circ - 40^\circ$ (1); $= 140^\circ$ (1); It equals $\angle A + \angle B = 35^\circ + 105^\circ$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 40°	Base angles are equal and the three angles add up to 180°.
2	**(d)** 37	Add the three sides.
3	**(c)** ASA	Compare the matching parts.
4	**(b)** 38°	The two acute angles add up to 90°.
5	**(c)** No	Two sides together must be longer than the third.
6	**(d)** 70°	The three angles add up to 180°.
7	**(a)** 60°	Recall the definition.
8	**(a)** 109°	Exterior angle = sum of the two interior opposite angles.
9	**(a)** 30°	Base angles are equal and the three angles add up to 180°.
10	**(d)** 24	Add the three sides.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle C = 180^\circ - 110^\circ = 70^\circ$; $\angle ACD = 180^\circ - 70^\circ$ (1); $= 110^\circ$ (equal to $\angle A + \angle B$) (1).	1+1
12	Two shorter sides: $4 + 6 = 10$; Compare with the longest side 10 (1); No, 10 is not greater than 10 (1).	1+1
13	Third $= 180^\circ - (55^\circ + 55^\circ)$; $= 70^\circ$ (1); The triangle is acute-angled (1).	1+1
14	$\angle C = 180^\circ - 105^\circ = 75^\circ$; $\angle ACD = 180^\circ - 75^\circ$ (1); $= 105^\circ$ (equal to $\angle A + \angle B$) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total parts $= 10$; one part $= 180^\circ \div 10 = 18^\circ$ (1); Angles 36°, 54°, 90° (1); Check: 180° (1).	1+1+1
16	Base angles are equal; their sum $= 180^\circ - 20^\circ = 160^\circ$ (1); Each base angle $= 80^\circ$ (1); Check: the three angles add up to 180° (1).	1+1+1
17	Exterior = sum of interior opposite angles (1); Other $= 150^\circ - 70^\circ = 80^\circ$ (1); Adjacent interior angle $= 180^\circ - 150^\circ = 30^\circ$ (1).	1+1+1
	OR: Total parts $= 9$; one part $= 180^\circ \div 9 = 20^\circ$ (1); Angles 20°, 60°, 100° (1); Check: 180° (1).	1+1+1
18	Both base angles are 35°, sum $= 70^\circ$ (1); Apex angle $= 180^\circ - 70^\circ$ (1); $= 110^\circ$ (1).	1+1+1
	OR: Exterior = sum of interior opposite angles (1); Other $= 125^\circ - 52^\circ = 73^\circ$ (1); Adjacent interior angle $= 180^\circ - 125^\circ = 55^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$10 + 10 = 20 > 15$, so the triangle exists (1); It is isosceles (1); Perimeter $= 10 + 10 + 15$ (1); $= 35$ cm (1); Final statement (1).	1+1+1+1+1
	OR: $\angle C = 180^\circ - 60^\circ - 60^\circ = 60^\circ$ (1); Classification: acute-angled (1); Exterior angle at C $= 180^\circ - 60^\circ$ (1); $= 120^\circ$ (1); It equals $\angle A + \angle B = 60^\circ + 60^\circ$ (1).	1+1+1+1+1
20	$4 + 7 = 11 > 9$, so the triangle exists (1); It is scalene (1); Perimeter $= 4 + 7 + 9$ (1); $= 20$ cm (1); Final statement (1).	1+1+1+1+1
	OR: $\angle C = 180^\circ - 25^\circ - 45^\circ = 110^\circ$ (1); Classification: obtuse-angled (1); Exterior angle at C $= 180^\circ - 110^\circ$ (1); $= 70^\circ$ (1); It equals $\angle A + \angle B = 25^\circ + 45^\circ$ (1).	1+1+1+1+1

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